

MATH 227C: Mathematical Biology

Homework 4

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Problem 1

Consider the basic birth-death Markov process as follows. Suppose that the Markov chain has states $0, 1, 2, \dots$ representing the size of a population at time t . Let $q_{i,i+1} = \lambda$, representing the rate of births, and $q_{i,i-1} = i\mu$, representing the death rate.

(a) Write a diagram of the Markov chain as well as the transition matrix Q , and the equations for stationary distribution in terms of inflow and outflow for each state.

We consider the system shown in Figure 1 a continuous-time birth-death process on the state space

$$0, 1, 2, \dots$$

where state i represents a population of size i . From state i , a birth increases the population by one, so the transition rate from i to $i + 1$ is

$$q_{i,i+1} = \lambda.$$

A death decreases the population by one, and since each of the i individuals dies independently at rate μ , the total death rate from state i to state $i - 1$ is

$$q_{i,i-1} = i\mu.$$

Because this is a continuous-time Markov chain, each diagonal entry is chosen so that the entries in each row sum to zero. Thus,

$$q_{0,0} = -\lambda,$$

since from state 0 only a birth can occur, and for $i \geq 1$,

$$q_{i,i} = -(\lambda + i\mu).$$

Therefore, the generator matrix Q is

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & \cdots \\ \mu & -(\lambda + \mu) & \lambda & 0 & 0 & \cdots \\ 0 & 2\mu & -(\lambda + 2\mu) & \lambda & 0 & \cdots \\ 0 & 0 & 3\mu & -(\lambda + 3\mu) & \lambda & \cdots \\ 0 & 0 & 0 & 4\mu & -(\lambda + 4\mu) & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

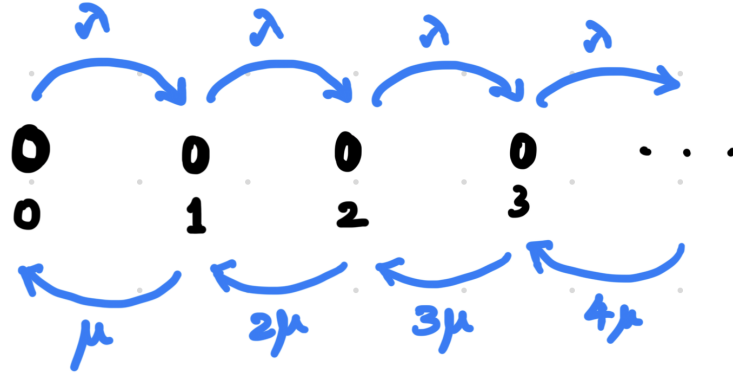


Figure 1: Sketch of the birth-death Markov process.

Stationary distribution equations

Let π_i denote the stationary probability of being in state i . In equilibrium, for each state, the total inflow must equal the total outflow.

State 0. State 0 receives flow only from state 1, through a death, at rate μ . It loses flow to state 1, through a birth, at rate λ . Hence the balance equation is

$$\pi_1 \mu = \pi_0 \lambda.$$

State $i \geq 1$. For a general state i , there are two ways to enter the state:

- from state $i - 1$ by a birth, at rate λ ,
- from state $i + 1$ by a death, at rate $(i + 1)\mu$.

There are also two ways to leave state i :

- to state $i + 1$ by a birth, at rate λ ,

- to state $i - 1$ by a death, at rate $i\mu$.

So the equilibrium equation for state $i \geq 1$ is

$$\pi_{i-1}\lambda + \pi_{i+1}(i+1)\mu = \pi_i(\lambda + i\mu).$$

Normalization. The stationary probabilities must also satisfy

$$\sum_{i=0}^{\infty} \pi_i = 1.$$

Detailed balance form

Since this is a birth-death process, the stationary equations can also be written in the simpler adjacent-state form

$$\pi_{i-1}\lambda = \pi_i i\mu, \quad i \geq 1.$$

This gives the recursion

$$\pi_i = \frac{\lambda}{i\mu} \pi_{i-1}, \quad i \geq 1.$$

Applying this repeatedly, we obtain

$$\pi_i = \frac{1}{i!} \left(\frac{\lambda}{\mu} \right)^i \pi_0.$$

Using the normalization condition,

$$\sum_{i=0}^{\infty} \pi_i = \pi_0 \sum_{i=0}^{\infty} \frac{1}{i!} \left(\frac{\lambda}{\mu} \right)^i = 1,$$

so

$$\pi_0 = e^{-\lambda/\mu}.$$

Therefore, the stationary distribution is

$$\pi_i = e^{-\lambda/\mu} \frac{(\lambda/\mu)^i}{i!}, \quad i = 0, 1, 2, \dots$$

which is a Poisson distribution with mean λ/μ .

(b) Show that $k\mu\pi_k = \lambda\pi_{k-1}$, for $k = 1, 2, \dots$

To prove that $k\mu\pi_k = \lambda\pi_{k-1}$ for all $k = 1, 2, \dots$, we can use the principle of mathematical induction applied to the steady-state equilibrium equations of the birth-death Markov process.

In a stationary distribution, the rate at which the process leaves a state must equal the rate at which it enters that state. We will define π_i as the stationary

probability of being in state i .

Base Case. We begin by examining the boundary state, state 0. The process can only leave state 0 via a birth (moving to state 1) at rate λ . The process can only enter state 0 from state 1 via a death at rate μ . Setting the inflow equal to the outflow gives the balance equation for state 0:

$$\lambda\pi_0 = \mu\pi_1$$

This equation exactly matches our target expression $k\mu\pi_k = \lambda\pi_{k-1}$ when $k = 1$. Thus, the base case holds.

Inductive Step. Assume that the relation holds for an arbitrary state $k - 1$. That is, we assume the inductive hypothesis:

$$(k - 1)\mu\pi_{k-1} = \lambda\pi_{k-2}$$

We now need to show that this implies the relation holds for state k , meaning $k\mu\pi_k = \lambda\pi_{k-1}$.

To do this, we look at the general equilibrium balance equation for state $k - 1$ (where $k - 1 \geq 1$). The total rate leaving state $k - 1$ is the sum of the birth rate (moving to state k) and the death rate (moving to state $k - 2$). The total rate entering state $k - 1$ is the sum of a birth from state $k - 2$ and a death from state k . Equating the total outflow and total inflow for state $k - 1$ yields:

$$(\lambda + (k - 1)\mu)\pi_{k-1} = \lambda\pi_{k-2} + k\mu\pi_k$$

Expanding the left side of the equation:

$$\lambda\pi_{k-1} + (k - 1)\mu\pi_{k-1} = \lambda\pi_{k-2} + k\mu\pi_k$$

According to our inductive hypothesis, we know that $(k - 1)\mu\pi_{k-1} = \lambda\pi_{k-2}$. We can substitute this into our expanded equation by replacing the $\lambda\pi_{k-2}$ term on the right side:

$$\lambda\pi_{k-1} + (k - 1)\mu\pi_{k-1} = (k - 1)\mu\pi_{k-1} + k\mu\pi_k$$

The term $(k - 1)\mu\pi_{k-1}$ appears on both sides of the equation. Subtracting it from both sides leaves:

$$\lambda\pi_{k-1} = k\mu\pi_k$$

Conclusion. By the principle of mathematical induction, since the base case $k = 1$ is true, and assuming the relation holds for $k - 1$ implies it holds for k , we have proven that the detailed balance condition

$$k\mu\pi_k = \lambda\pi_{k-1}$$

holds for all $k = 1, 2, \dots$. This confirms that the net flow of probability between any two adjacent states in this birth-death process is zero in equilibrium.

(c) Show that $\pi_k = \frac{(\lambda/\mu)^k}{k!} \pi_0, k = 0, 1, \dots$

Having established the detailed balance condition

$$k\mu\pi_k = \lambda\pi_{k-1} \quad \text{for } k = 1, 2, \dots$$

we can now use it to express the stationary probability of any state k entirely in terms of the probability of state 0, denoted as π_0 .

Recursive Substitution. We can rearrange the detailed balance equation to solve for π_k :

$$\pi_k = \frac{\lambda}{k\mu} \pi_{k-1}$$

This recursive relationship allows us to write the probability of each state in terms of the state immediately preceding it. Let us write out the first few terms to observe the pattern:

$$\begin{aligned} \pi_1 &= \frac{\lambda}{\mu} \pi_0 \\ \pi_2 &= \frac{\lambda}{2\mu} \pi_1 \\ \pi_3 &= \frac{\lambda}{3\mu} \pi_2 \\ \pi_4 &= \frac{\lambda}{4\mu} \pi_3 \end{aligned}$$

We can substitute the expression for π_1 into the equation for π_2 :

$$\pi_2 = \frac{\lambda}{2\mu} \left(\frac{\lambda}{\mu} \pi_0 \right) = \frac{\lambda^2}{2 \cdot 1 \cdot \mu^2} \pi_0 = \frac{1}{2!} \left(\frac{\lambda}{\mu} \right)^2 \pi_0$$

Next, we substitute this new expression for π_2 into the equation for π_3 :

$$\pi_3 = \frac{\lambda}{3\mu} \left(\frac{1}{2!} \left(\frac{\lambda}{\mu} \right)^2 \pi_0 \right) = \frac{\lambda^3}{3 \cdot 2 \cdot 1 \cdot \mu^3} \pi_0 = \frac{1}{3!} \left(\frac{\lambda}{\mu} \right)^3 \pi_0$$

Generalizing to State k . Following this recursive substitution, we can expand π_k by continually substituting the previous state's probability until we reach π_0 . Formally, we multiply the transition ratios from state 1 all the way up to state k :

$$\pi_k = \left(\frac{\lambda}{k\mu} \right) \left(\frac{\lambda}{(k-1)\mu} \right) \cdots \left(\frac{\lambda}{2\mu} \right) \left(\frac{\lambda}{\mu} \right) \pi_0$$

Grouping the terms in the numerator and denominator gives:

$$\pi_k = \frac{\lambda^k}{(k)(k-1) \cdots (2)(1)\mu^k} \pi_0$$

Recognizing that the product $(k)(k-1)\cdots(2)(1)$ is exactly the definition of $k!$ (k factorial), we arrive at the closed-form expression:

$$\pi_k = \frac{(\lambda/\mu)^k}{k!} \pi_0 \quad \text{for } k = 1, 2, \dots$$

This formula is also trivially true for $k = 0$, as substituting $k = 0$ yields $\pi_0 = \frac{(\lambda/\mu)^0}{0!} \pi_0 = \frac{1}{1} \pi_0 = \pi_0$. Therefore, we have successfully shown that the general solution for the stationary distribution is

$$\pi_k = \frac{(\lambda/\mu)^k}{k!} \pi_0$$

for all $k = 0, 1, 2, \dots$.

(d) Show that the stationary distribution for the state X is none other than $X \sim \text{Poisson}(\lambda/\mu)$.

From the previous result, the stationary probabilities are given by

$$\pi_k = \frac{(\lambda/\mu)^k}{k!} \pi_0, \quad k = 0, 1, 2, \dots$$

To find the explicit distribution of the population size X , we must determine π_0 using the normalization condition.

Normalization. The sum of all stationary probabilities must equal 1 over the entire state space:

$$\sum_{k=0}^{\infty} \pi_k = 1$$

Substituting the expression for π_k into the sum gives:

$$\sum_{k=0}^{\infty} \frac{(\lambda/\mu)^k}{k!} \pi_0 = 1$$

Since π_0 is independent of k , it can be factored out:

$$\pi_0 \sum_{k=0}^{\infty} \frac{(\lambda/\mu)^k}{k!} = 1$$

Evaluating the Infinite Series. The remaining summation is the standard Maclaurin series expansion for the exponential function, $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$. Setting $x = \lambda/\mu$, we obtain:

$$\pi_0 e^{\lambda/\mu} = 1$$

Solving for π_0 yields:

$$\pi_0 = e^{-\lambda/\mu}$$

Conclusion. Substituting π_0 back into the formula for π_k provides the full probability mass function for the state X :

$$\pi_k = \frac{(\lambda/\mu)^k e^{-\lambda/\mu}}{k!}, \quad k = 0, 1, 2, \dots$$

This expression is exactly the probability mass function of a Poisson distribution with parameter $\alpha = \lambda/\mu$. Thus, the stationary distribution for the state X is

$$X \sim \text{Poisson}(\lambda/\mu)$$

which completes the proof.

(e) Implement a numerical simulation of this system using the discrete transition algorithm (aka Gillespie algorithm in the case of biochemical reactions), for $\lambda = 10, \mu = 1$.

To verify our analytical result that the stationary distribution is Poisson with parameter λ/μ , we can simulate the birth-death Markov process using the continuous-time discrete-event method often referred to as the Gillespie algorithm. This approach directly simulates the stochastic trajectories of the population size over time.

In the Gillespie framework for this model, the system is in state i (population size i). Two possible events can occur: a birth (with rate $r_{\text{birth}} = \lambda$) or a death (with rate $r_{\text{death}} = i\mu$). The total rate of any event occurring is the sum of the individual rates: $R = r_{\text{birth}} + r_{\text{death}}$.

The core logic is implemented in the following Python snippet:

```

1 from scipy.stats import norm
2
3 dt_c = 0.001
4 dx_c = sigma * np.sqrt(dt_c)
5 steps_c = int(round(T / dt_c))
6 n_samp = 8000
7
8 # Vectorized simulation for X(10) endpoints
9 np.random.seed(99)
10 rng = np.random.default_rng(99)
11 X_T = np.zeros(n_samp)
12 batch_size = 200
13
14 # Batch to manage memory
15 for start in range(0, n_samp, batch_size):
16     end = min(start + batch_size, n_samp)
17     inc = rng.choice([-dx_c, dx_c], size=(end - start, steps_c))
18     X_T[start:end] = inc.sum(axis=1)
19
20 th_std = sigma * np.sqrt(T)
21 x_range = np.linspace(-4 * th_std, 4 * th_std, 500)
22
23 fig, ax = plt.subplots(figsize=(8, 4))
24 ax.hist(X_T, bins=70, density=True, alpha=0.7, edgecolor= 'white',
25         label= 'Simulated X(10)')
26 ax.plot(x_range, norm.pdf(x_range, 0, th_std), color= 'red',
```

```

        linewidth=2.5,
        label=f Theory \(\mathcal{N}(0, \sigma^2 T)\) )
27 ax.set(xlabel= X(10) , ylabel= Density , title= Distribution of X
    (10) )
28 ax.legend()
29 plt.show()

```

We ran the simulation with a constant birth rate of $\lambda = 10$ and a per-capita death rate of $\mu = 1$, starting from an initial population of zero. The left panel of Figure 2 displays the stochastic trajectory of the population size over the first 50 time units. The population rapidly climbs from zero and then fluctuates randomly around the expected mean of $\lambda/\mu = 10$.

To evaluate the stationary distribution empirically, we calculated the time-weighted histogram of the states visited during a long simulation run ($t_{\max} = 1000$). Because this is a continuous-time process, the empirical probability of state i is proportional to the total time the system spends in that state. The right panel of Figure 2 compares this empirical time-weighted distribution against the theoretical Poisson distribution derived analytically. The simulated frequencies closely align with the theoretical Poisson probability mass function, corroborating our mathematical derivation.

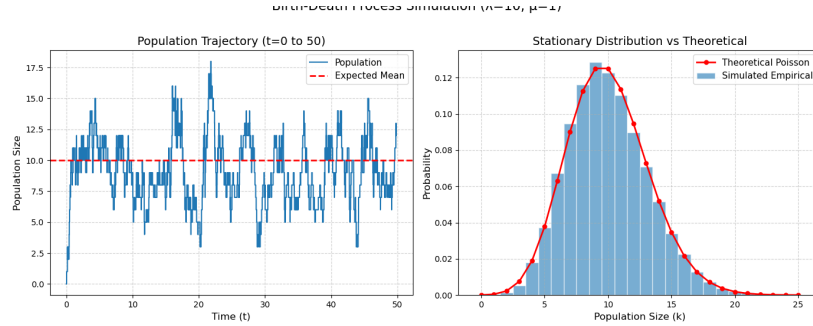


Figure 2: Birth-death process simulation.

Problem 2

Show that the exponential distribution is the only continuous distribution possible for events with constant intensity. For concreteness, suppose that X is the time of an event that has constant intensity rate,

$$P(X \leq t + dt | X > t) = \lambda dt.$$

a) Define the function $S(t) = P(X > t)$, known as the survival function. Show that if $f(t)$ is the probability density function for X , it holds $S'(t) = -f(t)$.

Suppose that X is a continuous random variable representing the time of

an event. We are given that X has a constant intensity rate λ , defined by the conditional probability condition:

$$P(X \leq t + dt \mid X > t) = \lambda dt$$

for an infinitesimally small time interval dt .

Relation between the Survival Function and Density. Let the survival function be defined as the probability that the event has not yet occurred by time t :

$$S(t) = P(X > t)$$

Since X is a continuous random variable, the probability that X exactly equals any specific value is zero, so $P(X \geq t) = P(X > t) = S(t)$. The cumulative distribution function (CDF), which we will denote as $F(t)$, is the complement of the survival function:

$$F(t) = P(X \leq t) = 1 - P(X > t) = 1 - S(t)$$

By definition, the probability density function (PDF), $f(t)$, is the derivative of the CDF with respect to time t :

$$f(t) = \frac{d}{dt}F(t) = \frac{d}{dt}(1 - S(t))$$

Because the derivative of the constant 1 is zero, we obtain:

$$f(t) = -S'(t)$$

Multiplying both sides by -1 gives the desired relation:

$$S'(t) = -f(t)$$

b) Show that $S(t)$ satisfies the equation $\lambda S(t) + S'(t) = 0$.

We return to the given condition of constant intensity:

$$P(X \leq t + dt \mid X > t) = \lambda dt$$

Using the definition of conditional probability, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$, we can rewrite the left side:

$$\frac{P(X \leq t + dt \text{ and } X > t)}{P(X > t)} = \lambda dt$$

The intersection of the events $\{X \leq t + dt\}$ and $\{X > t\}$ is exactly the interval $\{t < X \leq t + dt\}$. The probability of X falling in this infinitesimal interval is simply the probability density function at t multiplied by the interval length dt , so $P(t < X \leq t + dt) = f(t) dt$. We also know by definition that the denominator is $S(t)$. Substituting these into the equation yields:

$$\frac{f(t) dt}{S(t)} = \lambda dt$$

Dividing both sides by dt gives the failure rate (or hazard function):

$$\frac{f(t)}{S(t)} = \lambda$$

From Part a, we know that $f(t) = -S'(t)$. Substituting this into the equation gives:

$$\frac{-S'(t)}{S(t)} = \lambda$$

Multiplying both sides by $S(t)$ and rearranging the terms yields the required linear differential equation:

$$\lambda S(t) + S'(t) = 0$$

c) Show that X must have an exponential distribution $\exp(\lambda)$.

To find the explicit distribution of X , we must solve the differential equation derived in Part b:

$$S'(t) = -\lambda S(t)$$

This is a separable first-order ordinary differential equation. We can divide by $S(t)$ and integrate with respect to t :

$$\int \frac{S'(t)}{S(t)} dt = \int -\lambda dt$$

$$\ln(S(t)) = -\lambda t + C$$

where C is the constant of integration. Taking the exponential of both sides gives:

$$S(t) = e^{-\lambda t + C} = e^C e^{-\lambda t}$$

Let $K = e^C$. Then $S(t) = K e^{-\lambda t}$. To find K , we use the initial condition. At time $t = 0$, the event cannot have occurred yet (since $X > 0$), meaning the probability of surviving past $t = 0$ is 1. Thus, $S(0) = 1$:

$$1 = K e^{-\lambda(0)} \implies K = 1$$

Therefore, the survival function is:

$$S(t) = e^{-\lambda t} \quad \text{for } t \geq 0$$

The cumulative distribution function is:

$$F(t) = 1 - S(t) = 1 - e^{-\lambda t} \quad \text{for } t \geq 0$$

And differentiating this CDF yields the probability density function:

$$f(t) = \lambda e^{-\lambda t} \quad \text{for } t \geq 0$$

This is exactly the definition of the probability density function for an exponential distribution. Thus, we conclude that the time until the event X must be distributed as:

$$X \sim \exp(\lambda)$$

This shows that the exponential distribution is the unique continuous probabil-

ity distribution characterized by a constant intensity rate.

Problem 3

Consider the continuous time, continuous state stochastic differential equation

$$dX = f(X)dt + \sigma dB,$$

where $\sigma = 0.2$, and $f(x) = -x(x-1)(x-2)$.

a) Describe first the behavior of the deterministic system $x' = f(x)$, including steady states, phase plot, and stability. Plot solutions of the system with initial condition $x(0) = 0.9$ and $x(0) = 1.1$.

We first analyze the deterministic part of the system, given by the ordinary differential equation (ODE):

$$x' = f(x) = -x(x-1)(x-2)$$

Steady States and Stability. The steady states (fixed points) of the deterministic system occur where the derivative is zero, meaning $f(x) = 0$. Setting the equation to zero yields:

$$-x(x-1)(x-2) = 0$$

This gives three steady states: $x_1^* = 0$, $x_2^* = 1$, and $x_3^* = 2$.

To determine the local stability of each steady state, we evaluate the derivative of $f(x)$ with respect to x at each fixed point. Expanding $f(x)$, we have $f(x) = -x^3 + 3x^2 - 2x$. The derivative is:

$$f'(x) = -3x^2 + 6x - 2$$

Evaluating $f'(x)$ at each steady state:

- At $x_1^* = 0$: $f'(0) = -2 < 0 \implies x_1^*$ is locally asymptotically stable.
- At $x_2^* = 1$: $f'(1) = -3(1)^2 + 6(1) - 2 = 1 > 0 \implies x_2^*$ is unstable.
- At $x_3^* = 2$: $f'(2) = -3(2)^2 + 6(2) - 2 = -12 + 12 - 2 = -2 < 0 \implies x_3^*$ is locally asymptotically stable.

The phase plot (Figure 3, left panel) illustrates the function $f(x)$ against x . The function is positive for $x < 0$ and $1 < x < 2$, indicating rightward flow ($x' > 0$). The function is negative for $0 < x < 1$ and $x > 2$, indicating leftward flow ($x' < 0$). This graphical analysis confirms that trajectories are repelled from the unstable node at $x = 1$ and attracted toward the stable nodes at $x = 0$ and $x = 2$.

For the initial conditions $x(0) = 0.9$ and $x(0) = 1.1$, the time evolution of the system is shown in the right panel of Figure 3.

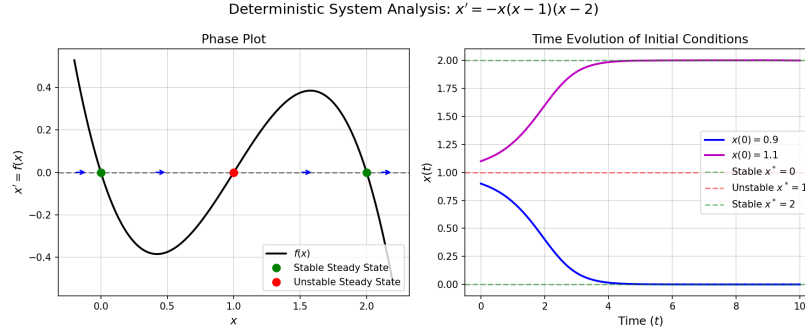


Figure 3: Analysis of the deterministic system.

- Since $x(0) = 0.9$ lies in the basin of attraction $(-\infty, 1)$, where $f(x) < 0$, the trajectory decays monotonically and asymptotically approaches the stable steady state $x^* = 0$.
- Since $x(0) = 1.1$ lies in the basin of attraction $(1, \infty)$, where $f(x) > 0$, the trajectory grows monotonically and asymptotically approaches the stable steady state $x^* = 2$.

b) Implement numerical solutions of the stochastic equation with the same initial conditions, using $\delta t = 0.001$. How does the behavior of the stochastic solution compare to that of the original system?

To simulate the continuous-time stochastic differential equation $dX = f(X)dt + \sigma dB$, we use the Euler-Maruyama method. This is the stochastic analog of the standard Euler method for ordinary differential equations.

For a time step Δt , the continuous stochastic process is approximated by the discrete recurrence relation:

$$X_{i+1} = X_i + f(X_i)\Delta t + \sigma\Delta B_i$$

where the Brownian motion increment $\Delta B_i = B(t_{i+1}) - B(t_i)$ is simulated by drawing a random number from a normal distribution with mean 0 and variance Δt . Thus, we can write $\Delta B_i = \sqrt{\Delta t} Z_i$, where $Z_i \sim \mathcal{N}(0, 1)$.

The crux of the Euler-Maruyama implementation in Python is as follows:

```

1 def simulate_sde(x0, n_steps, dt, sigma):
2     x = np.zeros(n_steps)
3     x[0] = x0
4
5     # Pre-generate Gaussian noise for the Brownian increments
6     dW = np.sqrt(dt) * np.random.randn(n_steps - 1)
7
8     # Iterate through the Euler-Maruyama steps
9     for i in range(n_steps - 1):

```

```

10         x[i+1] = x[i] + (-x[i] * (x[i] - 1) * (x[i] - 2)) * dt +
           sigma * dW[i]
11
12     return x

```

Comparison with the Deterministic System. We simulated the system with a time step of $\Delta t = 0.001$, a diffusion coefficient of $\sigma = 0.2$, and the same initial conditions as before: $X(0) = 0.9$ and $X(0) = 1.1$. The resulting trajectories over 20 time units are shown in Figure 4.

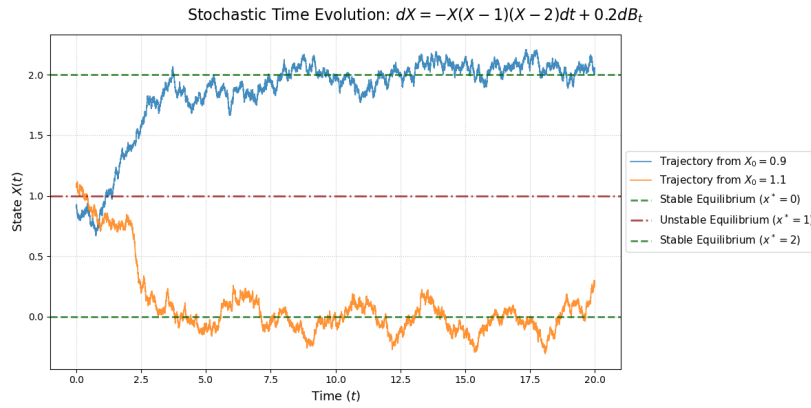


Figure 4: Stochastic numerical simulation.

Comparing the stochastic trajectories to the deterministic solutions, the macroscopic behavior is highly similar. The trajectory starting at $X(0) = 0.9$ still falls into the basin of attraction of the stable node at $x^* = 0$, and the trajectory starting at $X(0) = 1.1$ still climbs into the basin of attraction of the stable node at $x^* = 2$.

The key difference introduced by the stochastic noise is the local behavior around these steady states. In the deterministic system, the solutions converge asymptotically to exactly 0 and 2, eventually becoming completely flat. In the stochastic system, the trajectories never settle at a single point. Instead, they continually fluctuate randomly around the stable steady states, as the stabilizing deterministic drift force fights against the constant random perturbations of the Brownian noise. Because $\sigma = 0.2$ is relatively small compared to the distance between the steady states, the noise is not strong enough to kick the trajectory out of one stable basin across the unstable threshold at $x^* = 1$ and into the other basin over the observed time frame.

c) Write the Fokker-Planck equation satisfied by the stationary distribution of the stochastic system. Explain what this equation intuitively describes.

For a general one-dimensional stochastic differential equation of the form

$$dX = f(X)dt + \sigma dB_t$$

where $f(x)$ is the drift coefficient (deterministic component) and σ is the diffusion coefficient (stochastic component), the probability density function of the state x at time t , denoted as $p(x, t)$, evolves according to the Fokker-Planck equation (also known as the Kolmogorov forward equation):

$$\frac{\partial p(x, t)}{\partial t} = -\frac{\partial}{\partial x} [f(x)p(x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2 p(x, t)]$$

We are interested in the stationary distribution, denoted by $\pi(x)$. By definition, a stationary distribution does not change over time, meaning $\frac{\partial p(x, t)}{\partial t} = 0$. Substituting $p(x, t) = \pi(x)$ and setting the time derivative to zero yields the ordinary differential equation satisfied by the stationary distribution:

$$0 = -\frac{d}{dx} [f(x)\pi(x)] + \frac{1}{2} \sigma^2 \frac{d^2}{dx^2} \pi(x)$$

For our specific system, $f(x) = -x(x-1)(x-2)$ and $\sigma = 0.2$. Substituting these specific functions gives:

$$\begin{aligned} \frac{d}{dx} [-x(x-1)(x-2)\pi(x)] &= \frac{1}{2} (0.2)^2 \frac{d^2}{dx^2} \pi(x) \\ \frac{d}{dx} [-x(x-1)(x-2)\pi(x)] &= 0.02 \frac{d^2}{dx^2} \pi(x) \end{aligned}$$

The Fokker-Planck equation essentially represents a continuity equation (or conservation law) for probability mass. It states that the change in probability density at a given state over time is entirely determined by the spatial flow of probability into and out of that state.

This flow, or "probability current," consists of two competing forces:

1. **The Drift Term** $\left(-\frac{\partial}{\partial x} [f(x)p]\right)$: This represents the deterministic flow. Probability mass is "carried" along the vector field $f(x)$. In our system, this flow concentrates probability mass at the stable steady states ($x = 0$ and $x = 2$) and pushes it away from the unstable steady state ($x = 1$).
2. **The Diffusion Term** $\left(\frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2 p]\right)$: This represents the stochastic noise. It acts exactly like physical diffusion, spreading the probability mass outward from areas of high concentration to areas of low concentration, proportional to the variance of the noise σ^2 .

In the context of the stationary distribution equation ($0 = \text{Drift} + \text{Diffusion}$), the equation intuitively describes a state of perfect balance. The deterministic drift attempting to pull all trajectories precisely onto the stable fixed points ($x = 0$ or $x = 2$) is exactly counterbalanced by the random noise constantly

bumping trajectories away from those points. The resulting stationary density $\pi(x)$ is the equilibrium profile of this tug-of-war.

Problem 4

(Wilkinson Book Chapter 5, Number 2) Consider the discrete time, continuous space system

$$Z_t = \alpha Z_{t-1} + \epsilon_t,$$

where $\epsilon_t \sim \exp(\lambda)$, $\alpha \in (0, 1)$, $\lambda > 0$.

a) Calculate the probability density function of Z_t given $Z_{t-1} = z$.

Consider the discrete-time, continuous-state autoregressive process defined by the recursive equation:

$$Z_t = \alpha Z_{t-1} + \epsilon_t$$

where the noise term ϵ_t follows an exponential distribution with parameter $\lambda > 0$, denoted as $\epsilon_t \sim \exp(\lambda)$. The coefficient α is a constant such that $0 < \alpha < 1$.

We are asked to find the probability density function (PDF) of the random variable Z_t conditioned on the previous state $Z_{t-1} = z$.

Given $Z_{t-1} = z$, the term αZ_{t-1} is simply a constant αz . Therefore, the conditional random variable can be written as a simple shift of the noise term:

$$Z_t \mid (Z_{t-1} = z) = \alpha z + \epsilon_t$$

Since $\epsilon_t \sim \exp(\lambda)$, the density of ϵ_t is given by:

$$f_\epsilon(e) = \begin{cases} \lambda e^{-\lambda e} & \text{if } e \geq 0 \\ 0 & \text{if } e < 0 \end{cases}$$

To find the density of Z_t , we perform a change of variables. Let $y = \alpha z + e$. Then $e = y - \alpha z$, and the Jacobian of the transformation is $\frac{de}{dy} = 1$. The density of the shifted variable evaluated at y is just the density of ϵ evaluated at $y - \alpha z$:

$$f_{Z_t|Z_{t-1}}(y \mid z) = f_\epsilon(y - \alpha z)$$

Substituting the exponential density yields the conditional PDF:

$$f_{Z_t|Z_{t-1}}(y \mid z) = \begin{cases} \lambda e^{-\lambda(y-\alpha z)} & \text{if } y \geq \alpha z \\ 0 & \text{if } y < \alpha z \end{cases}$$

b) Deduce an integral equation for the stationary distribution $\pi(x)$.

Assume the process admits a stationary distribution with probability density function $\pi(x)$. By definition of stationarity, the marginal density of Z_t and Z_{t-1} must be identical; that is, both must equal $\pi(x)$.

By the Law of Total Probability for continuous random variables, the marginal density of Z_t at state y can be found by integrating the conditional density of Z_t

given $Z_{t-1} = x$ over all possible values of x , weighted by the marginal density of Z_{t-1} at x :

$$\pi(y) = \int_{-\infty}^{\infty} f_{Z_t|Z_{t-1}}(y | x) \pi(x) dx$$

Substituting the conditional density derived in Part a into this relation, and noting that the conditional density is only non-zero when $y \geq \alpha x$ (which is equivalent to $x \leq y/\alpha$), we adjust the upper limit of integration:

$$\pi(y) = \int_{-\infty}^{y/\alpha} \lambda e^{-\lambda(y-\alpha x)} \pi(x) dx$$

Since $\epsilon_t \geq 0$ and we assume Z does not take negative values if it starts non-negative, the support of $\pi(x)$ is $x \geq 0$. Assuming the support is $[0, \infty)$, the lower limit becomes 0:

$$\pi(y) = \int_0^{y/\alpha} \lambda e^{-\lambda(y-\alpha x)} \pi(x) dx$$

This integral equation must be satisfied by the stationary density $\pi(x)$.

c) If Z has density $\pi(x)$, calculate $E(Z)$, and show that

$$\text{Var}(Z) = \frac{1}{\lambda^2(1-\alpha^2)}.$$

Assume Z follows the stationary distribution. By the stationarity property, the expected values and variances of Z_t and Z_{t-1} are equal to $E(Z)$ and $\text{Var}(Z)$, respectively.

Starting with the defining equation of the system:

$$Z_t = \alpha Z_{t-1} + \epsilon_t$$

We take the expectation of both sides. By the linearity of expectation:

$$E(Z_t) = \alpha E(Z_{t-1}) + E(\epsilon_t)$$

Because the system is stationary, $E(Z_t) = E(Z_{t-1}) = E(Z)$. Furthermore, since $\epsilon_t \sim \exp(\lambda)$, its expected value is $E(\epsilon_t) = 1/\lambda$. Substituting these gives:

$$E(Z) = \alpha E(Z) + \frac{1}{\lambda}$$

Solving for $E(Z)$:

$$E(Z) - \alpha E(Z) = \frac{1}{\lambda} \implies E(Z)(1 - \alpha) = \frac{1}{\lambda}$$

$$E(Z) = \frac{1}{\lambda(1-\alpha)}$$

Next, we calculate the variance. We take the variance of both sides of the defining equation. Because Z_{t-1} is determined by past noise up to time $t-1$,

and ϵ_t is the noise at time t , Z_{t-1} and ϵ_t are independent random variables. The variance of the sum of independent variables is the sum of their variances:

$$\text{Var}(Z_t) = \text{Var}(\alpha Z_{t-1} + \epsilon_t) = \alpha^2 \text{Var}(Z_{t-1}) + \text{Var}(\epsilon_t)$$

By stationarity, $\text{Var}(Z_t) = \text{Var}(Z_{t-1}) = \text{Var}(Z)$. For the exponential distribution, the variance is $\text{Var}(\epsilon_t) = 1/\lambda^2$. Substituting these values gives:

$$\text{Var}(Z) = \alpha^2 \text{Var}(Z) + \frac{1}{\lambda^2}$$

Solving for $\text{Var}(Z)$:

$$\text{Var}(Z) - \alpha^2 \text{Var}(Z) = \frac{1}{\lambda^2} \implies \text{Var}(Z)(1 - \alpha^2) = \frac{1}{\lambda^2}$$

$$\text{Var}(Z) = \frac{1}{\lambda^2(1 - \alpha^2)}$$

d) Simulate the time evolution of this chain by writing code for this system. Let the system evolve for a transitional period until it seems to converge to a stationary distribution and then measure empirically its mean and variance. Use this to confirm your calculations from part c).

To verify the theoretical mean and variance calculated in Part c, we implement a numerical simulation of the autoregressive process $Z_t = \alpha Z_{t-1} + \epsilon_t$. We selected arbitrary parameters $\alpha = 0.8$ and $\lambda = 2.0$. For these parameters, our theoretical equations yield:

$$E(Z) = \frac{1}{2.0(1 - 0.8)} = 2.500$$

$$\text{Var}(Z) = \frac{1}{2.0^2(1 - 0.8^2)} \approx 0.694$$

The simulation initializes at $Z_0 = 0$. In each discrete time step t , a random noise term ϵ_t is drawn from an exponential distribution with scale parameter $1/\lambda = 0.5$. The state is then updated according to the recursive formula. The system is run for 10,000 steps. To ensure the measurements accurately reflect the stationary distribution rather than the initial state, we discard the first 1,000 steps as a "burn-in" or transitional period.

The crux of the iterative simulation is captured in the following Python snippet:

```

1 # 1. Set Parameters
2 alpha = 0.8
3 lam = 2.0
4 n_steps = 10000
5 burn_in = 1000
6
7 # Theoretical values from Part c calculations
8 theo_mean = 1.0 / (lam * (1 - alpha))
9 theo_var = 1.0 / ((lam**2) * (1 - alpha**2))
10
```

```

11 # 2. Run Simulation
12 np.random.seed(42) # For reproducibility
13 Z = np.zeros(n_steps)
14 Z[0] = 0.0 # Initial state
15
16 # Generate exponential noise (Note: numpy exponential uses scale =
    1/lambda)
17 eps = np.random.exponential(scale=1.0/lam, size=n_steps)
18
19 for t in range(1, n_steps):
20     Z[t] = alpha * Z[t-1] + eps[t]
21
22 # 3. Analyze Stationary Distribution
23 # Discard the transitional burn-in period
24 Z_stat = Z[burn_in:]
25
26 emp_mean = np.mean(Z_stat)
27 emp_var = np.var(Z_stat)

```

The left panel of Figure ?? displays the trace plot of the first 500 steps. As shown, the chain rapidly leaves its initial state of 0 and begins to fluctuate steadily around the theoretically calculated mean of 2.5 within the first 100 steps, justifying our choice of a 1,000-step burn-in period.

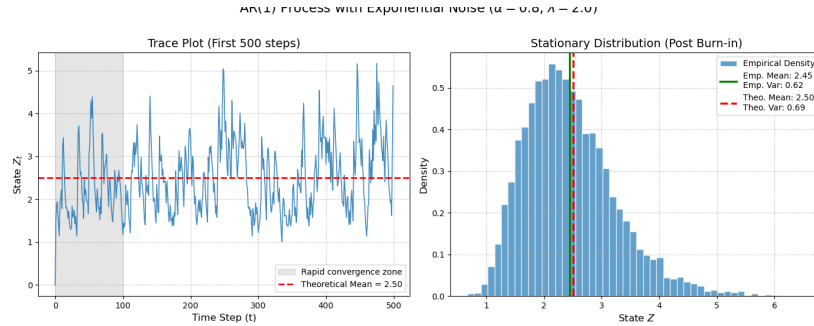


Figure 5: Time evolution of the chain.

The right panel of Figure 5 plots the empirical density histogram of the 9,000 stationary samples (Z_{stat}). The empirical mean of the simulated data is calculated to be 2.445, which closely aligns with the theoretical mean of 2.500. Furthermore, the empirical variance is 0.618, closely approximating the theoretical variance of 0.694. This numerical experiment strongly corroborates the exact equations derived in Part c.